The cytoskeleton

Overview

1) Microtubules (25 nm Ø)

2) Microfilaments (7 nm Ø)

3) Thick filaments (15 nm Ø)

4) Intermediate filaments (8-12 nm Ø)

Study material

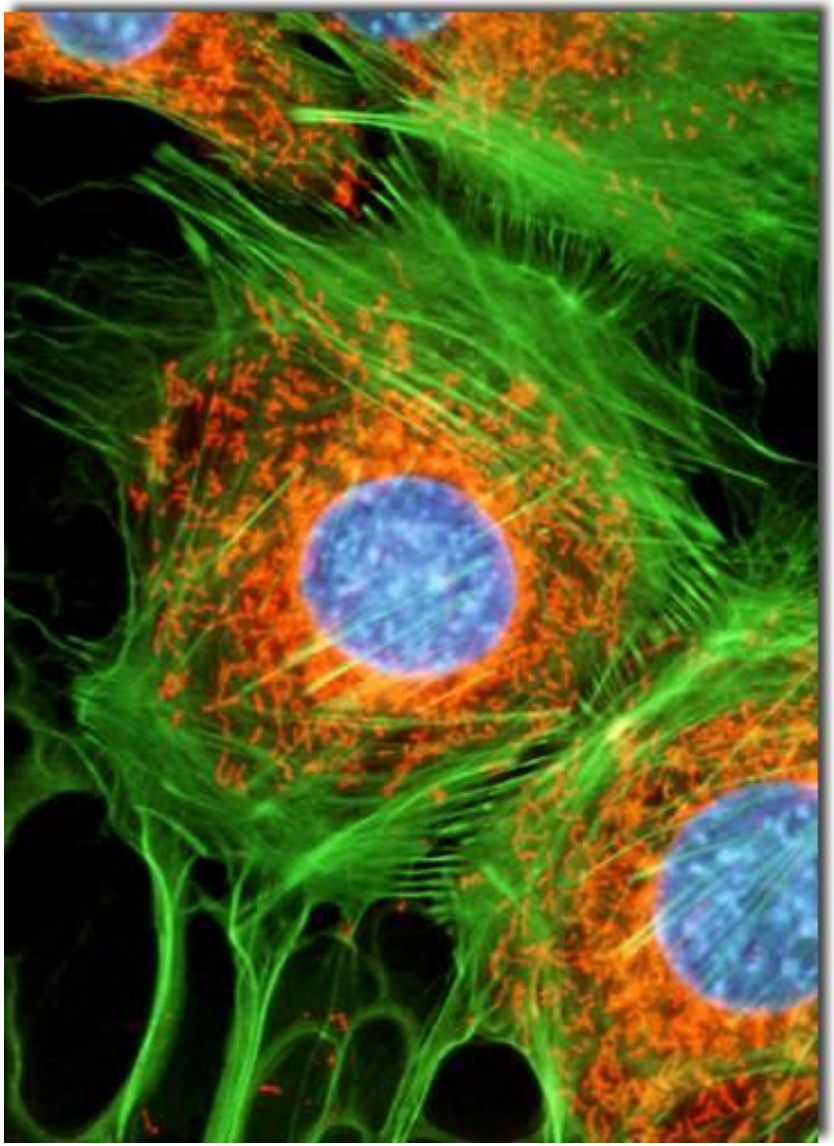
STUDY MATERIAL TO COMPLETE THE LEC. 3.7

SUMMARY MATERIAL FROM THE FINAL PART OF CHAPTER 12 OF THE BOOK BIOMEDICAL CELLULAR BIOLOGY: The cytoskeleton

Autthors: Alfonso Calvo, José J. Abad, Francisco Expósito, Luis Montuenga

Learning objectives:

- Know the structure and function of intermediate filaments.



The cytoskeleton

Overview

1) Microtubules (25 nm Ø)

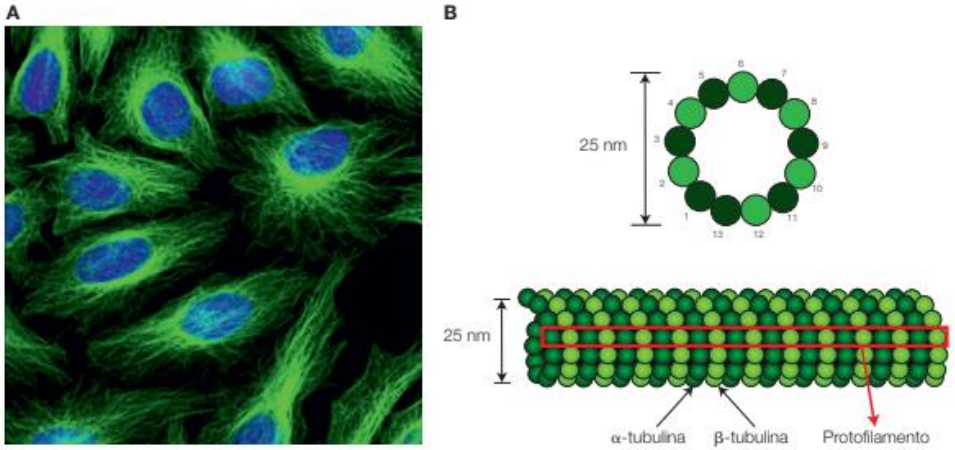
2) Microfilaments (7 nm Ø)

3) Thick filaments (15 nm Ø)

4) Intermediate filaments (8-12 nm Ø)

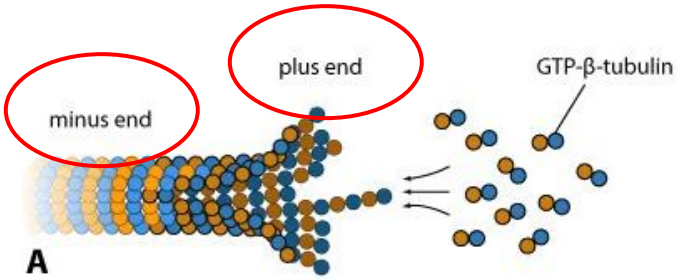


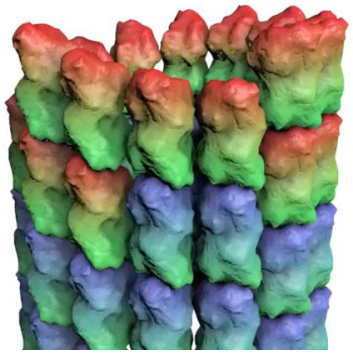
Microtubules



Made of dimers of α-β-tubulin

Important properties:
-Polarity
-Dynamic instability
-GTP-dependent

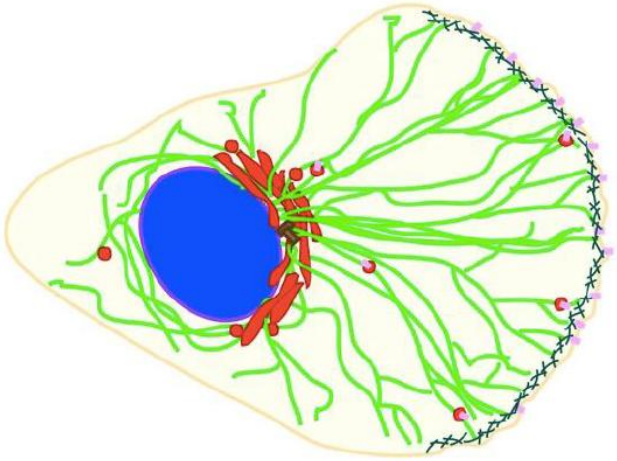


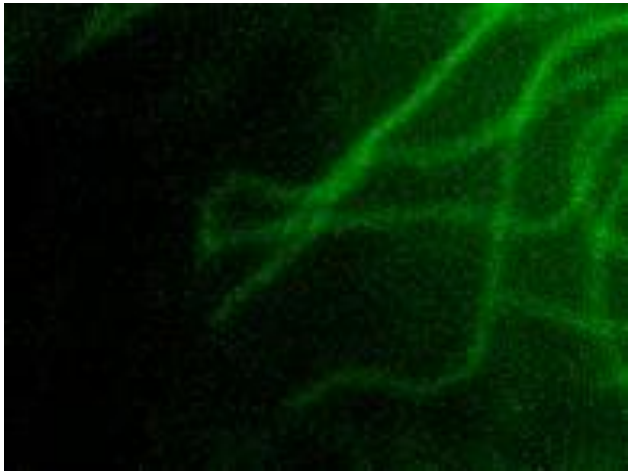


Dynamic instability: a constant process of microtubule growing and shortening.

Dynamic instability: to learn more about the mechanism of dynamic instability: see Biología Celular Biomédica, capítulo 12. **Not required for the exam.**

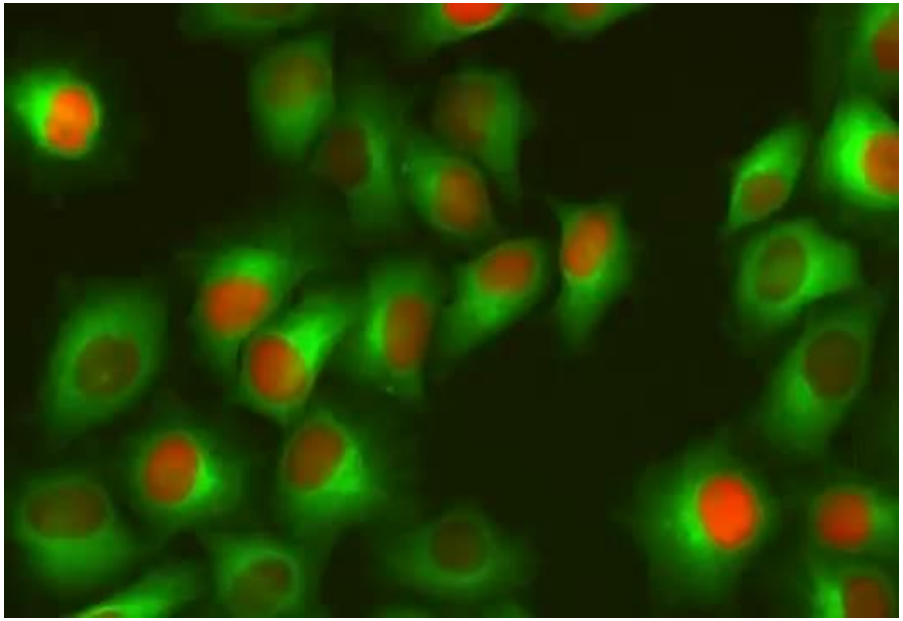
Microtubules also show dynamic instability controlled by different proteins





Functions of microtubules

1. Maintenance of cell shape and polarity.
2. Cell division (spindle formation, chromatid segregation).
3. Transport of vesicles and organelles
4. Cell movement.



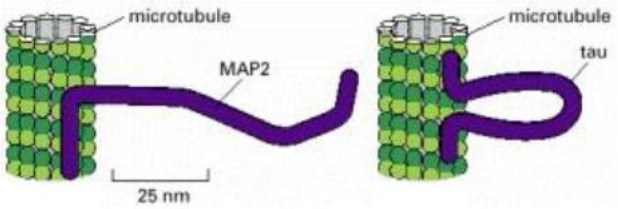
Microtubule associated proteins (MAPs)
(Examples)

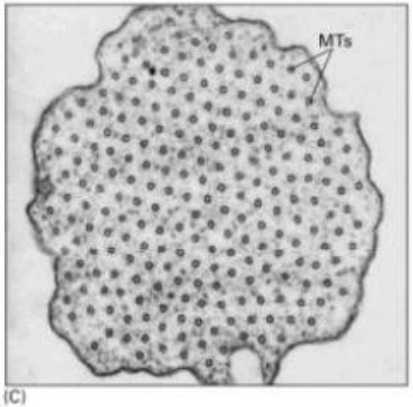
Non-motor MAPs

-***MAP2*** and ***Tau***

-Stabilization, cross-linking

-Alzheimer disease





Alteration of Tau in Alzheimer disease



https://www.youtube.com/watch?v=AV1BqpOyHmM

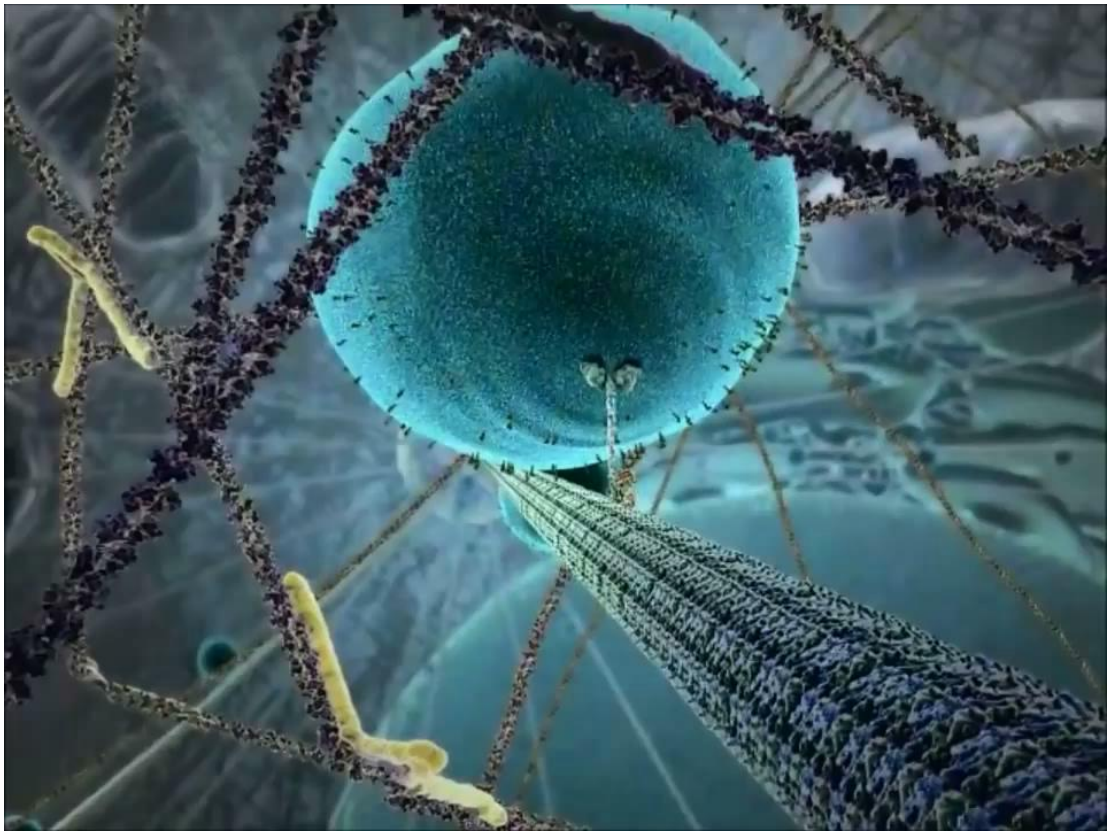
Microtubule associated proteins (MAPs)
(Examples)

Motor MAPs

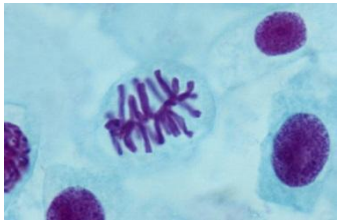
-***kinesin*** (-to +) and ***dynein*** (+ to -)

-Vesicle trafficking

-ATP-dependent movement

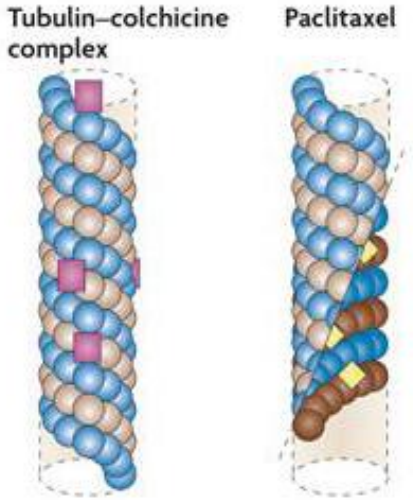


Agents that alter microtubule dynamics



AGENTS THAT BLOCK POLYMERIZATION

-Colchicine: used to prepare the karyotype



AGENTS THAT BLOCK DEPOLYMERIZATION

-Taxanes used to treat cancer

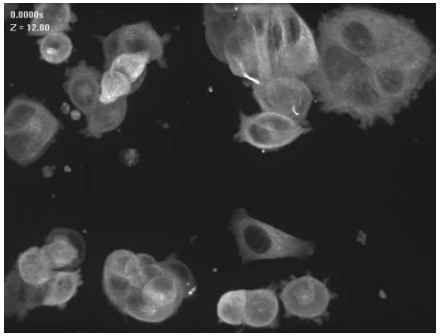




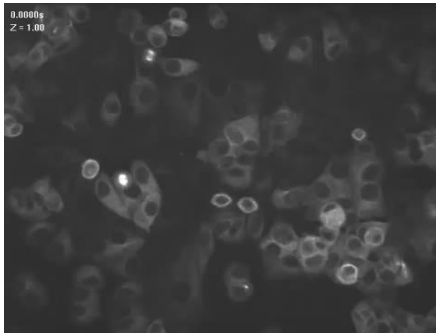
Yew tree

Effect of taxanes on breast cancer cells

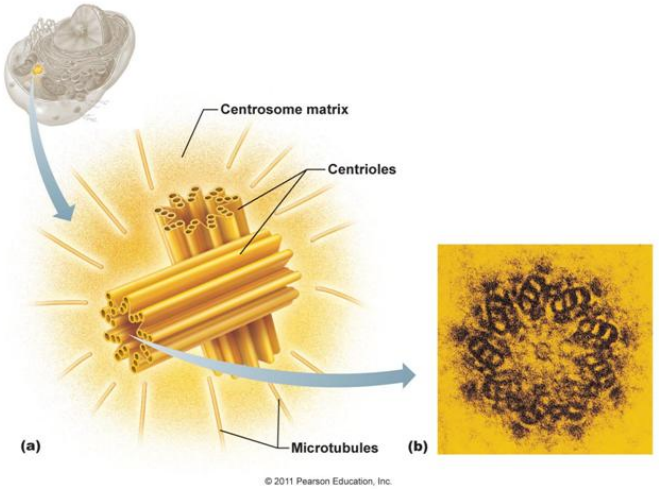
Untreated

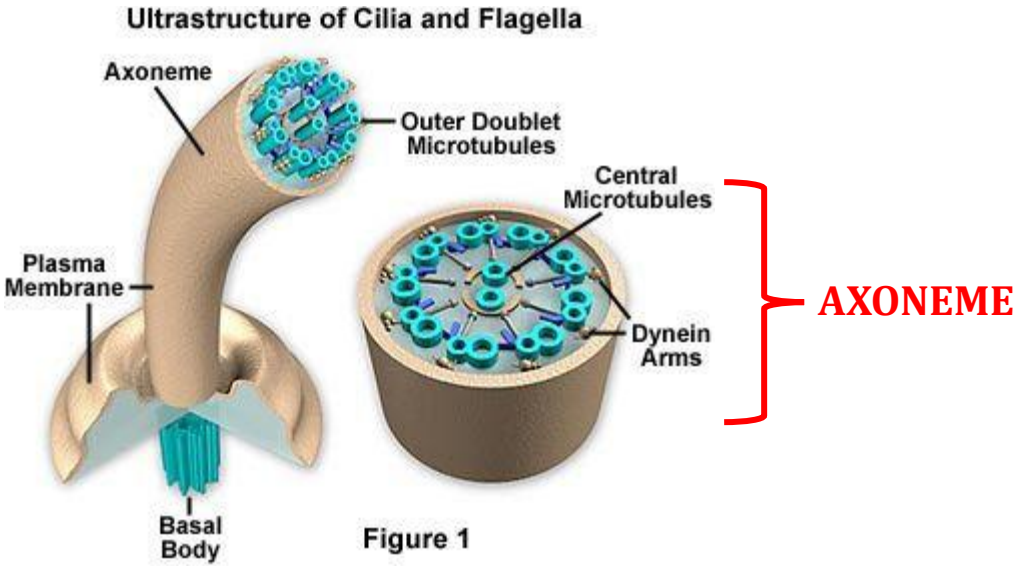


Treated



Centrioles and motile cilia





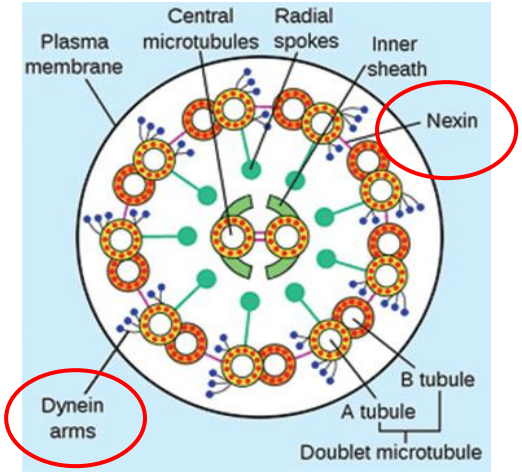
CENTRIOLES

-9_3+0 structure

-No membrane

-Localization:

–Centrosome (pair of centrioles)



MOTILE CILIA

Stem:

-9_2+2 structure

-Surrounded by membrane

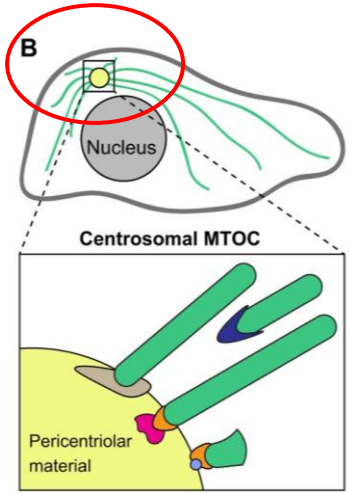
Basal body:

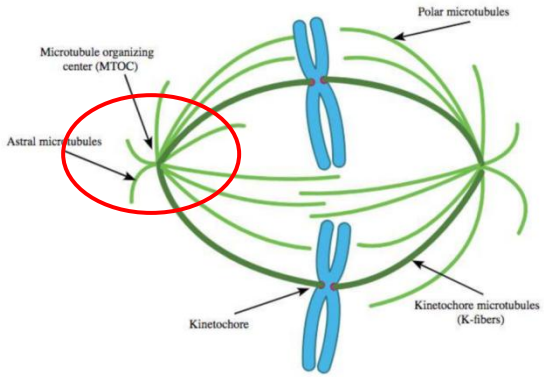
-9_3+0 structure

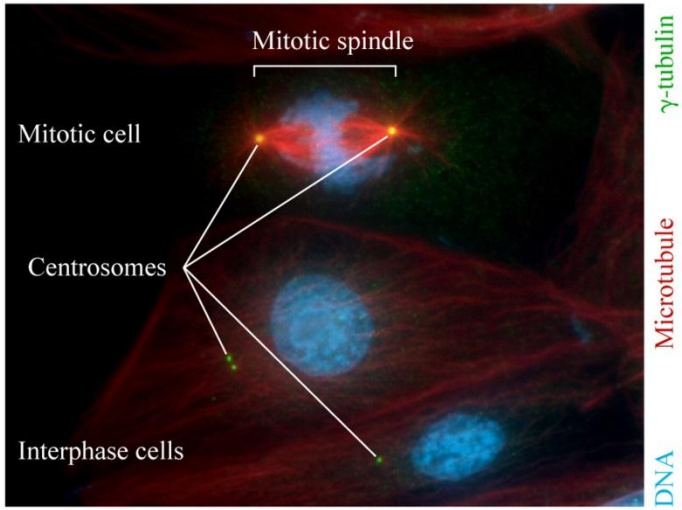
-Below the membrane

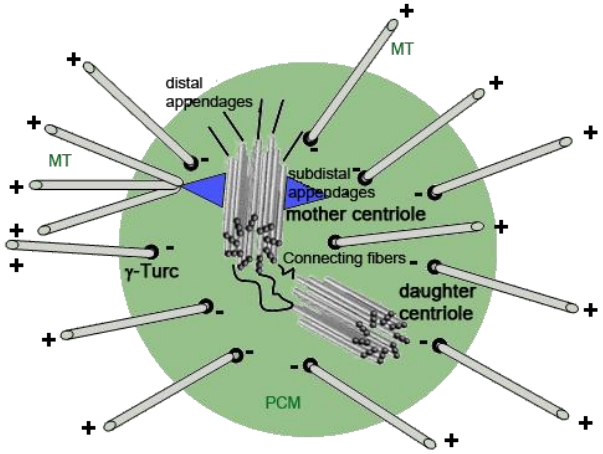
Microtubule organizing centers (MTOC)

MTOC are: centrioles and basal bodies of cilia





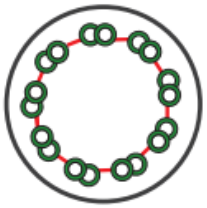


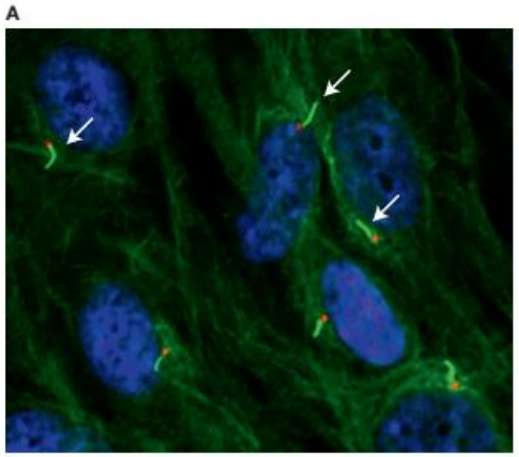


Gamma (γ) tubulin in MTOC is needed for microtubule growth

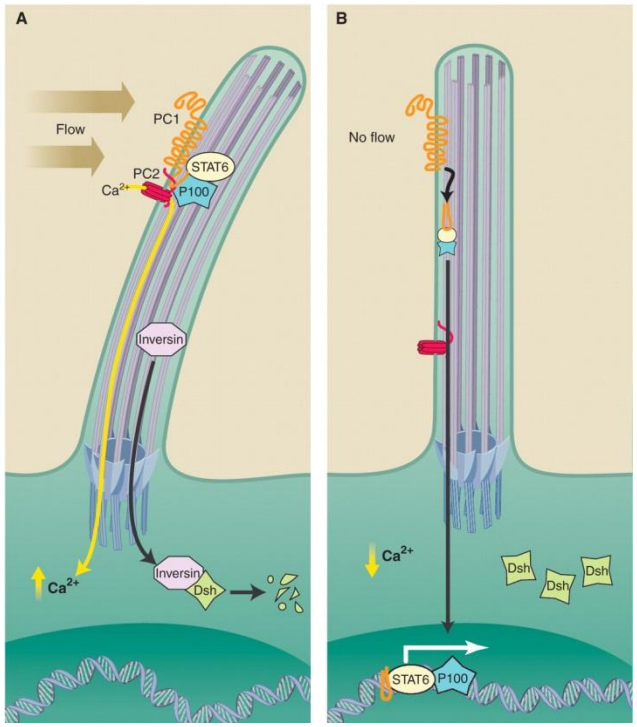
Immotile cilia

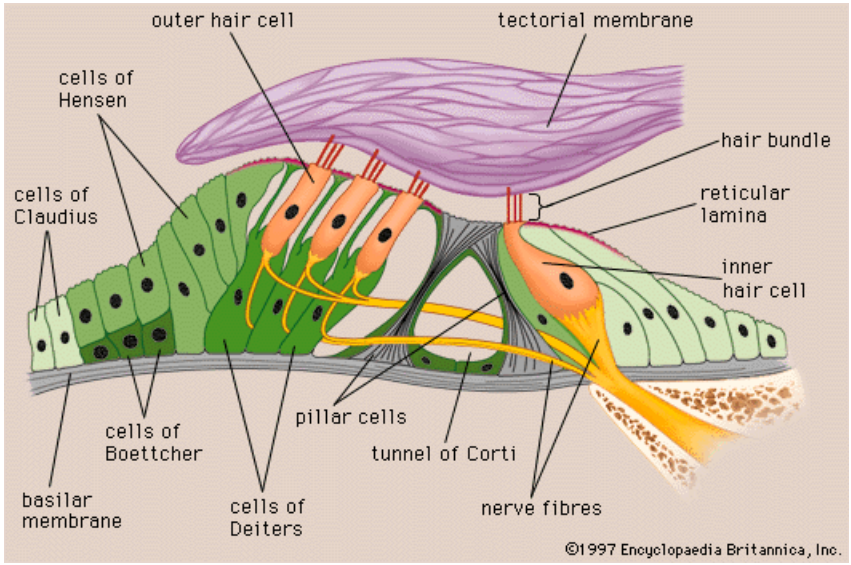
IMMOTILE CILIA
-9_2+0 structure
-Membrane
-Called primary cilia
-Function: Sensor (antennae)











Primary ciliary dyskinesia (PCD)

CHARACTERISTICS

-Rare autosomal recessive syndrome

-Axoneme alteration

-90%: **Dynein mutations**

-Cilia and flagella dysfunctions (lack of motility)

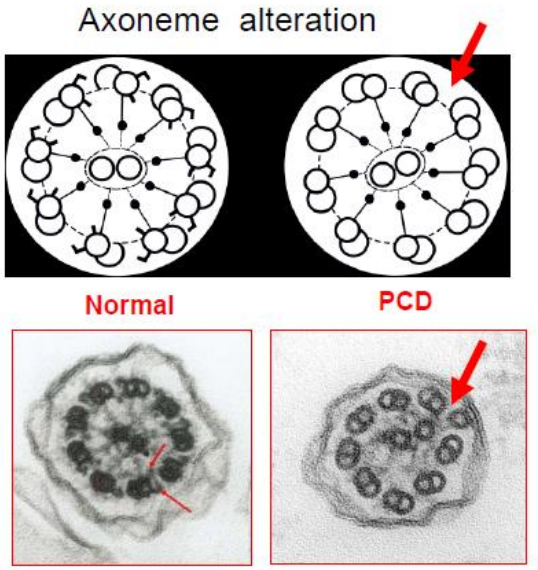
CLINICAL MANIFESTATIONS

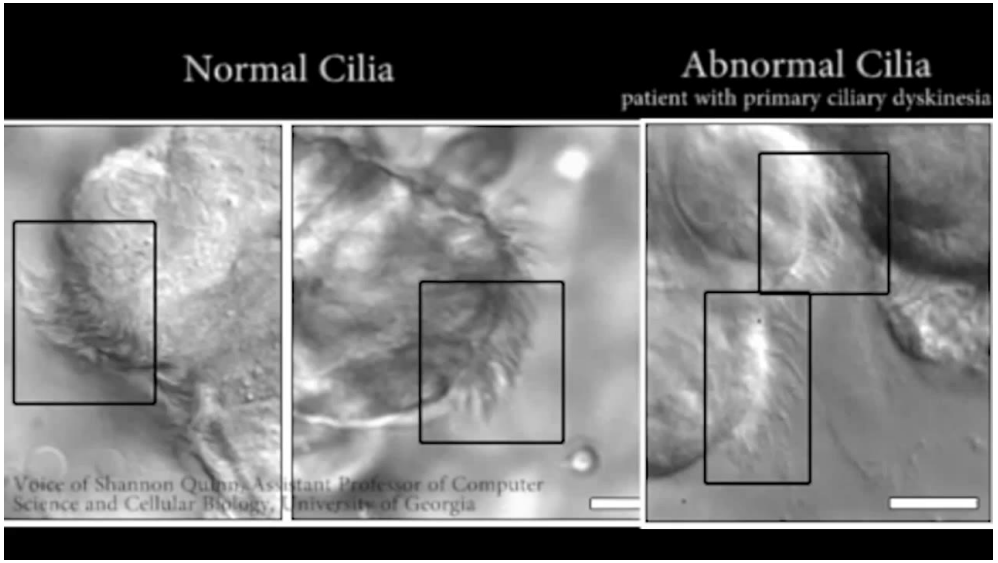
1. Chronic sinusitis, bronchitis, pneumonia

2. Infertility

-Lack of sperm motility

-Oviduct dysfunction



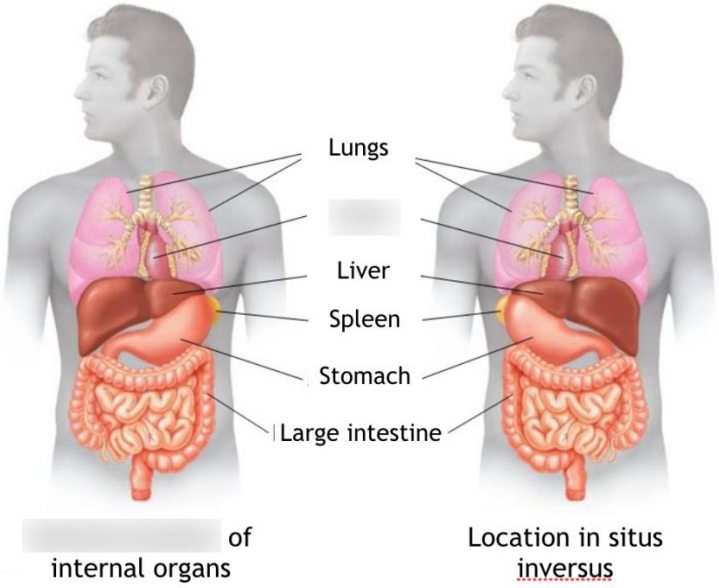


Situs inversus

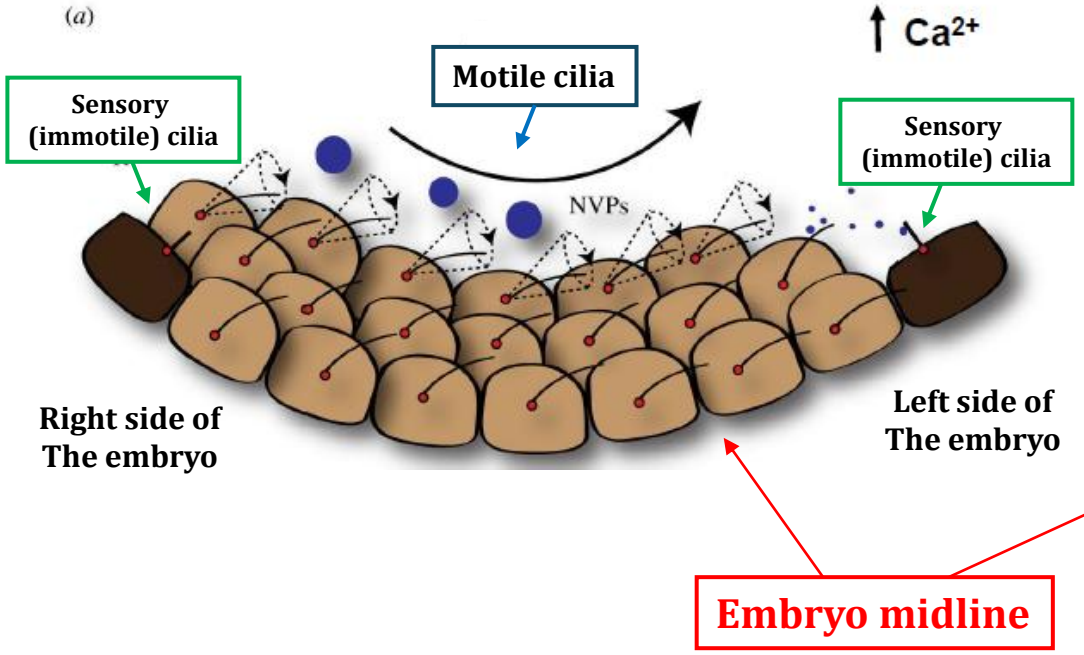
-Reversal location of some asymmetric organs

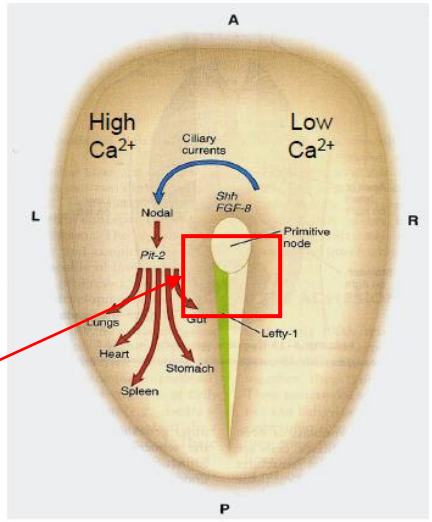
-95%: no symptoms or dysfunctions

-50% patients with **PCD** show situs inversus



How situs inversus is produced





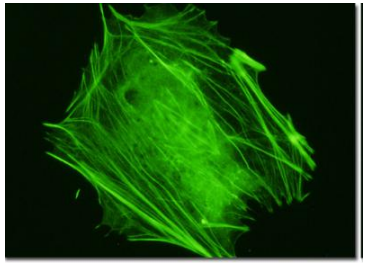
Microfilaments (actin)

-Made of actin monomers Actin { G (globular) actin (monomer); F (filamentous) actin (7nm Ø)

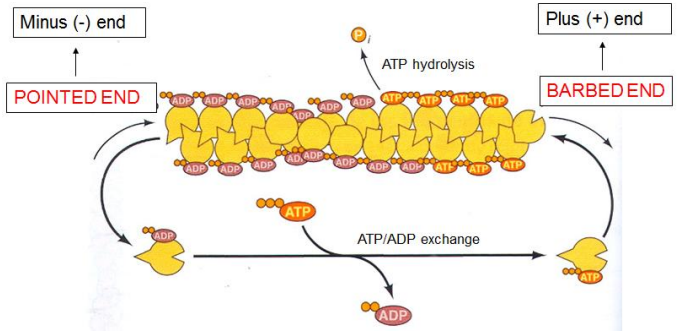
-Polarized and dynamic (<dynamic instability)

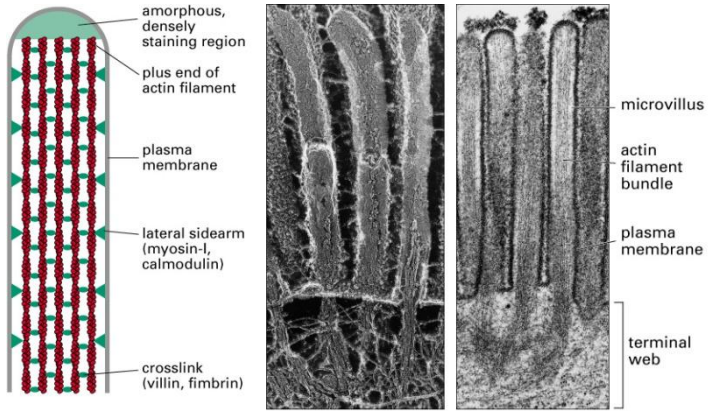
-ATP is needed for polymerization

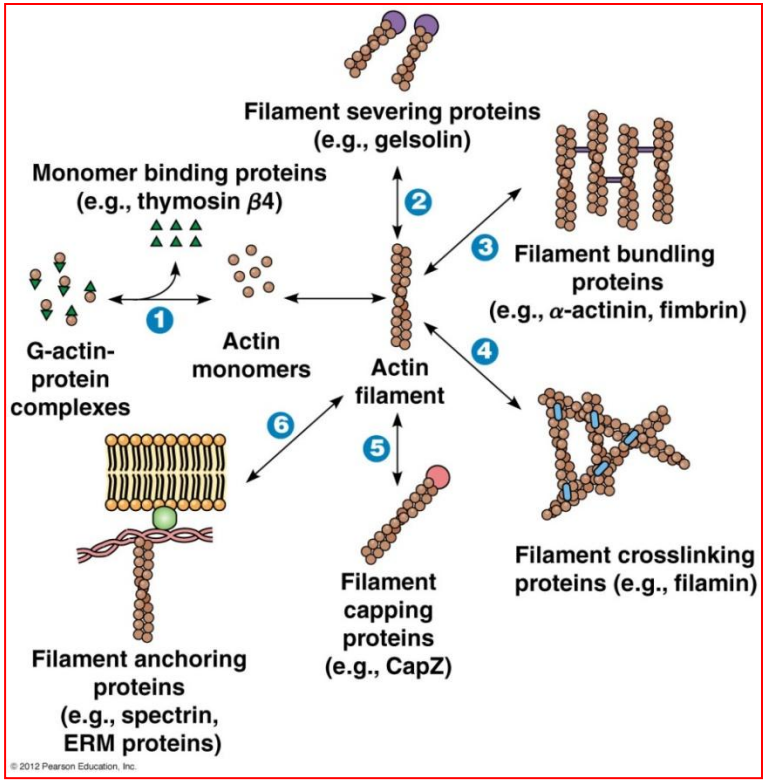
-Actin-binding proteins → microfilaments stability and association



Actin filaments





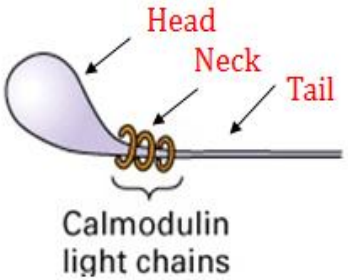


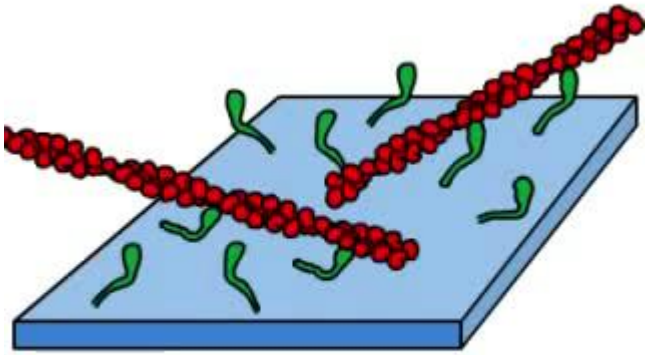
Do not learn these names

Myosin: motor protein that interacts with actin filaments

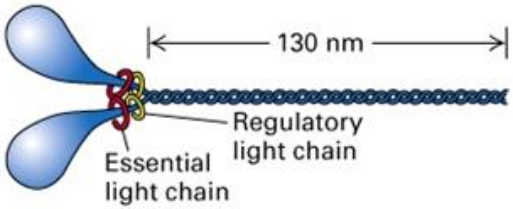
Head: ATPase activity ⟶ The head bends

Myosin I

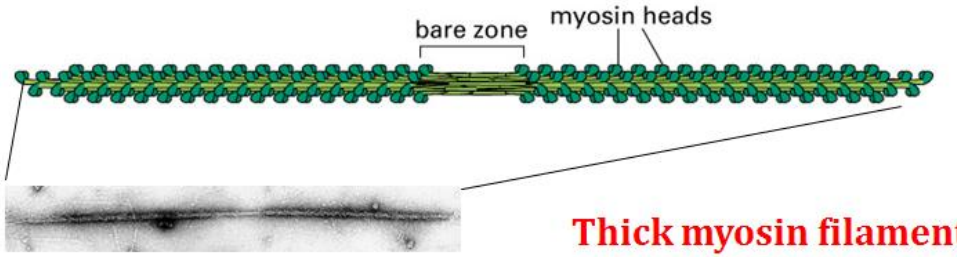




Myosin II





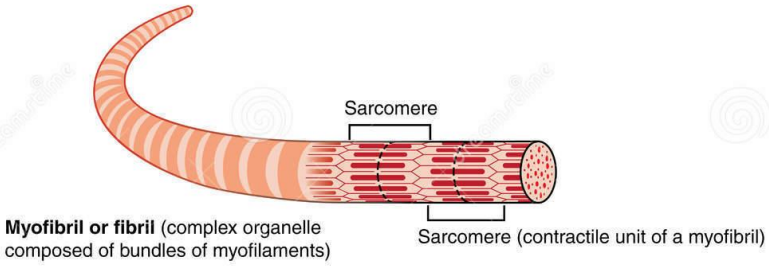


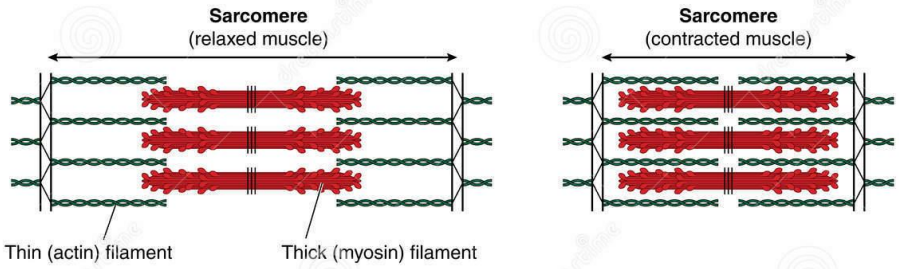
Thick myosin filament 15 nm Ø

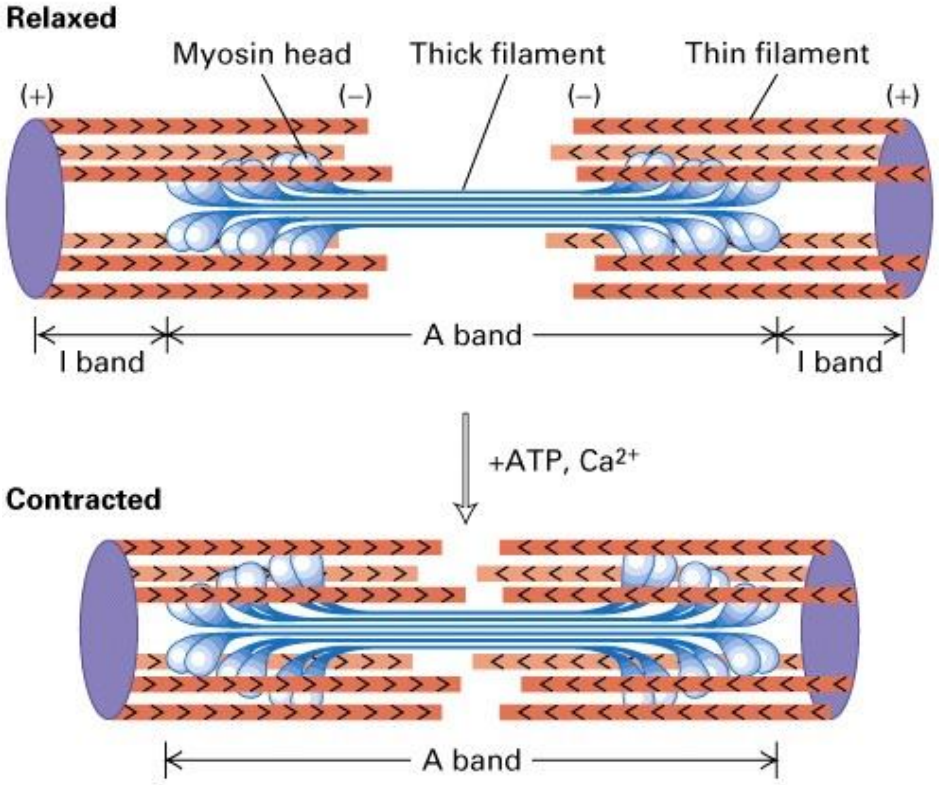
The sarcomere

(The functional unit or muscle contraction)



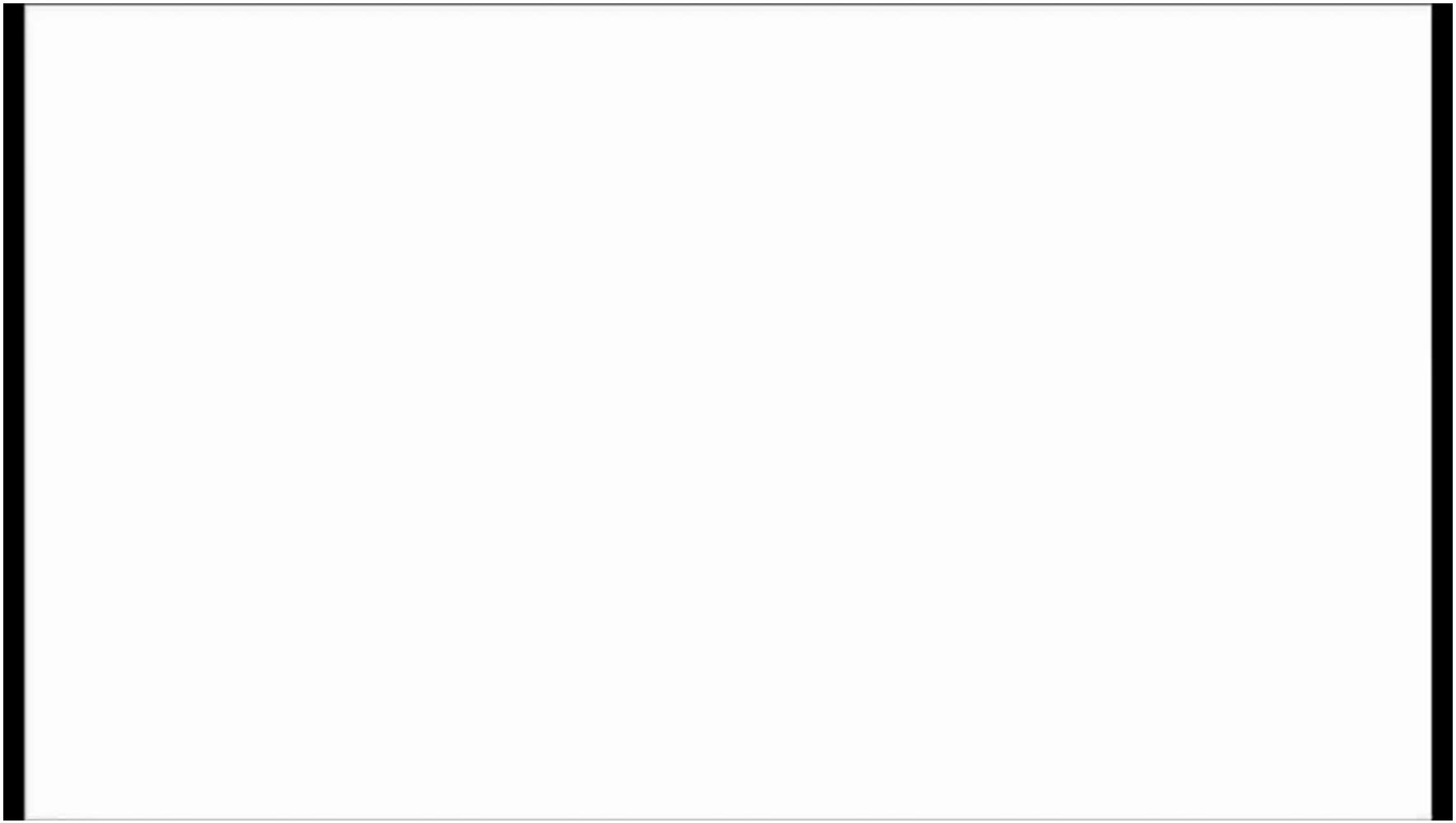




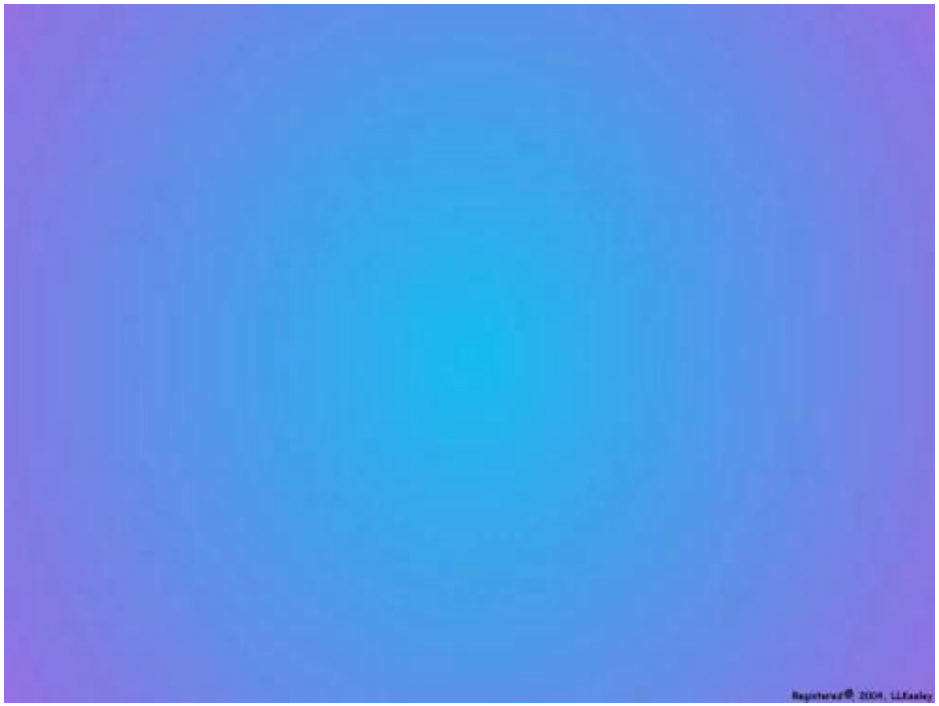


Molecular biology of muscle contraction video

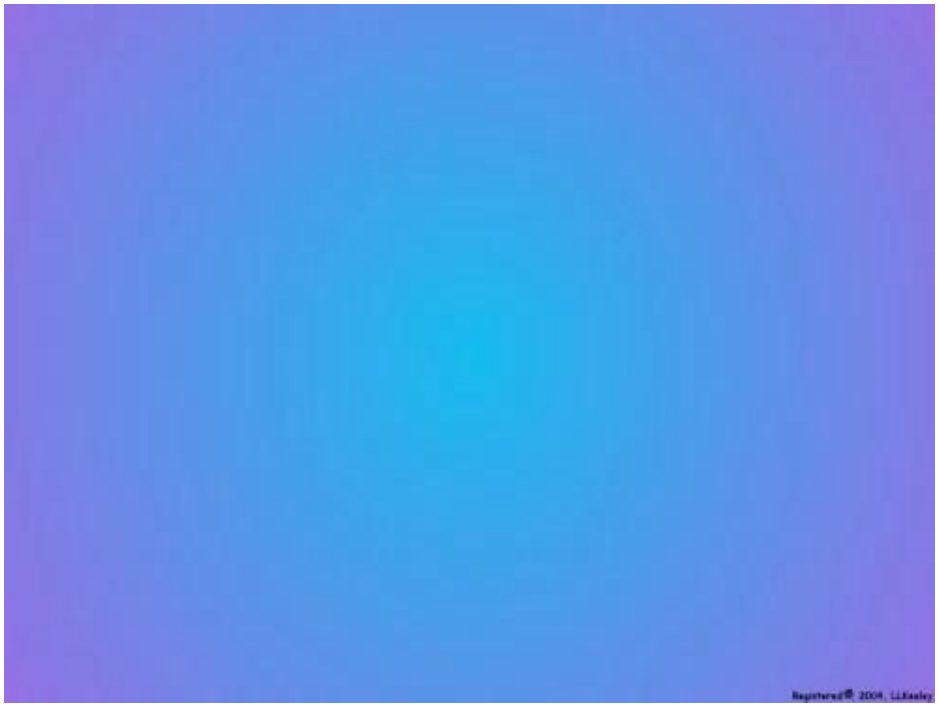
https://www.youtube.com/watch?v=nTZnBdeIb5c



Molecular biology of muscle contraction video



https://www.youtube.com/watch?v=J9K1enGBJRw



https://www.youtube.com/watch?v=4CMmG4T3s5k

Actin-myosin in non-muscle cells

Actin-myosin interactions allow cell crawling. **Not organized in sarcomeres**

Cell migration
- Embryogenesis
- Metastasis
- Inflammation

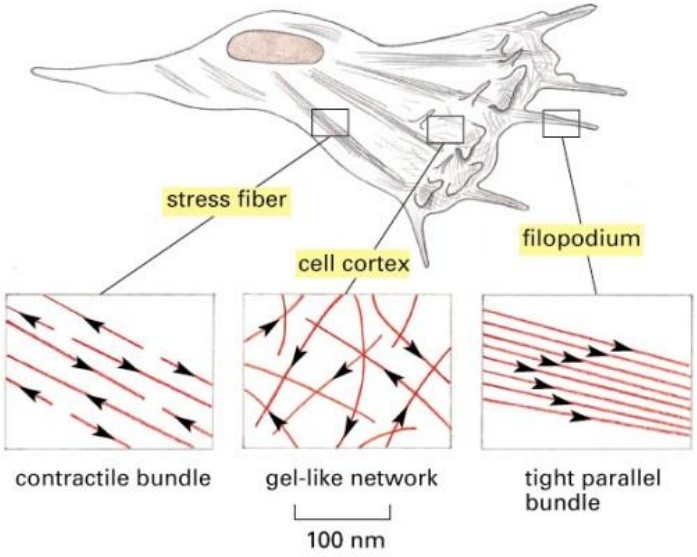


Figure 16–38. Molecular Biology of the Cell, 4th Edition.

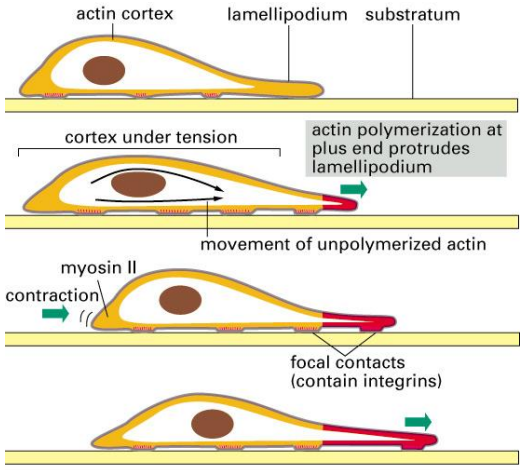
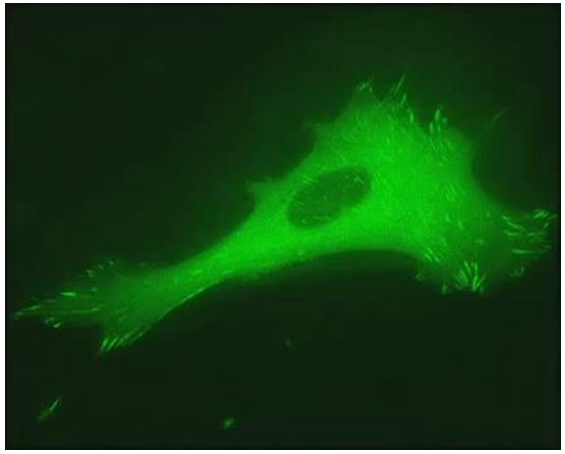


Figure 16–85. Molecular Biology of the Cell, 4th Edition.

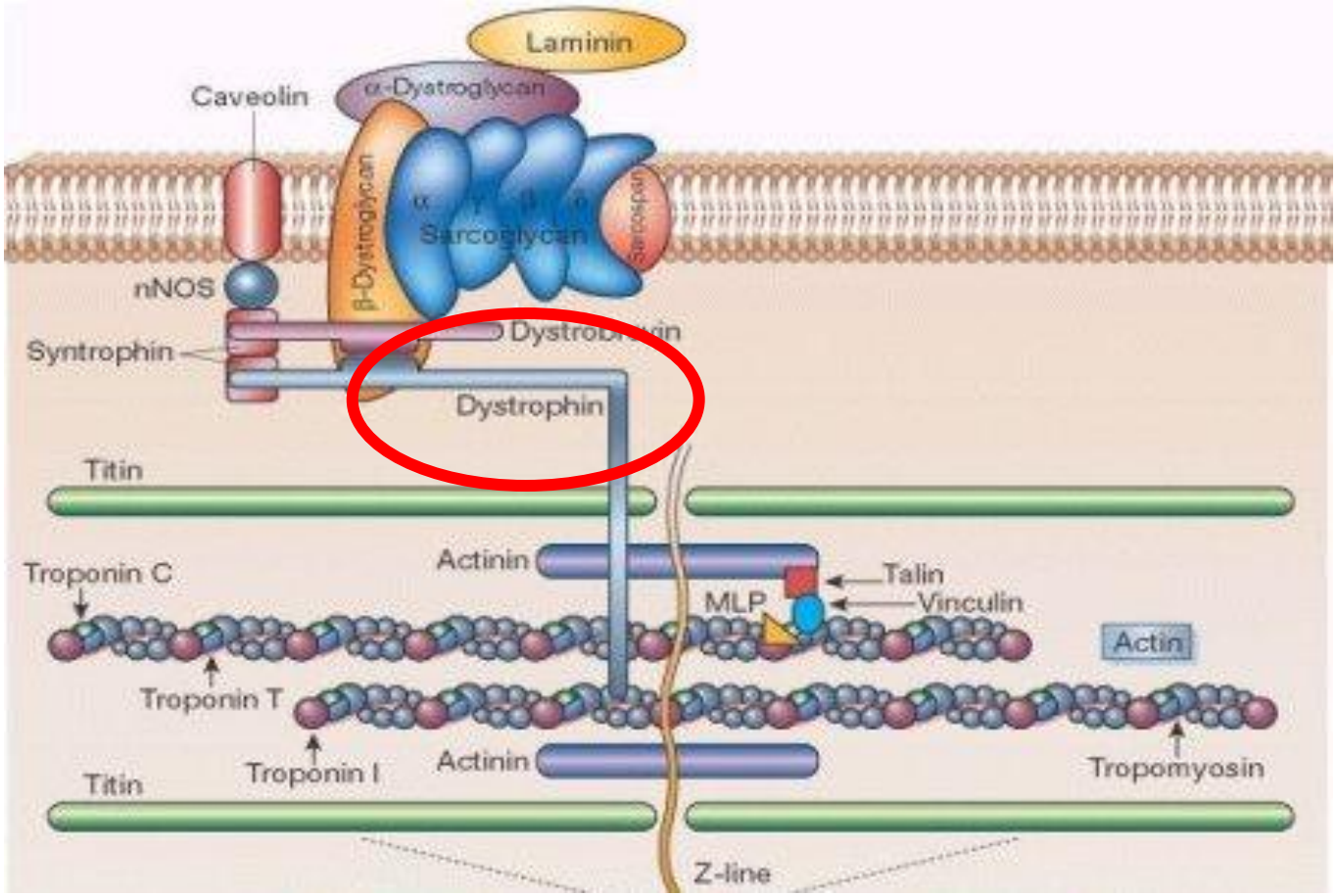
The crawling cell…



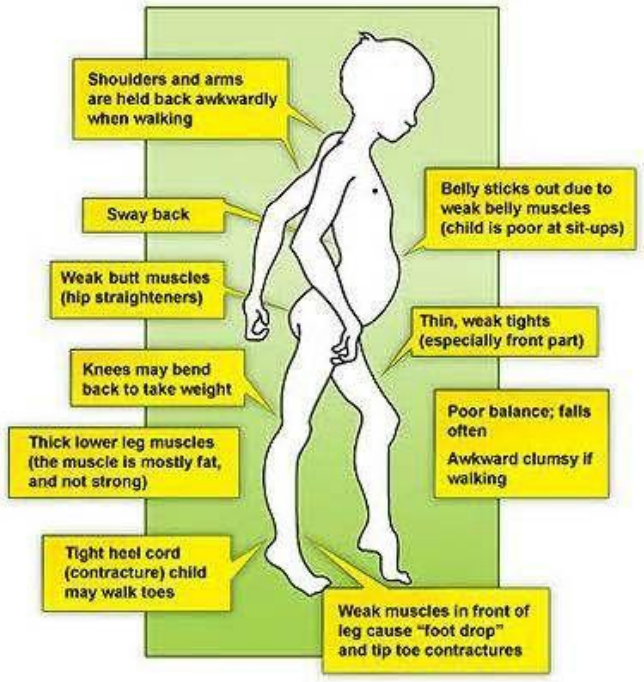
Duchenne's muscular dystrophy

- Genetic disorder **(X chromosome-linked)**: disorganization of the sarcomere

- Mutations in dystrophin (protein bound to actin in muscle cells)

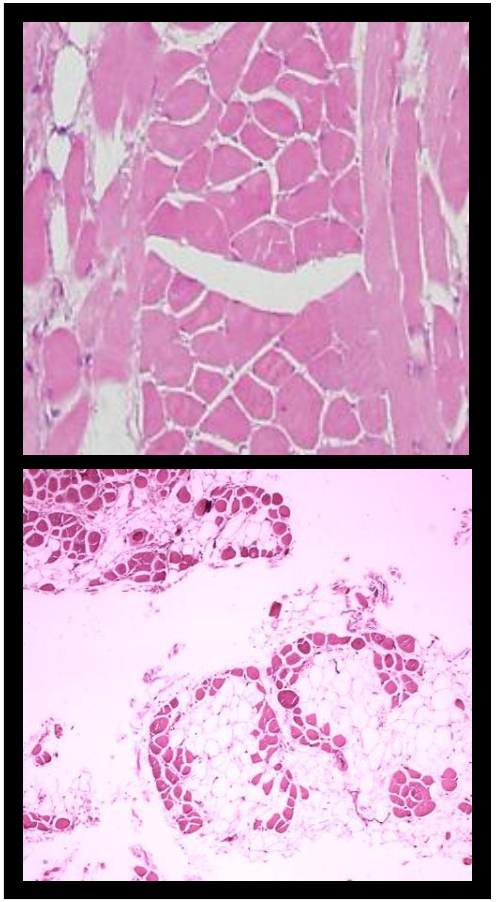


Progressive muscle degeneration



Normal

Dystrophy



CLINICAL SYMPTOMS

- Weakness
- Fatigue
- Motor difficulties
- Limb deformities
- Breathing problems
- Cardiomyopathy

Therapies for Duchenne's muscular dystrophy

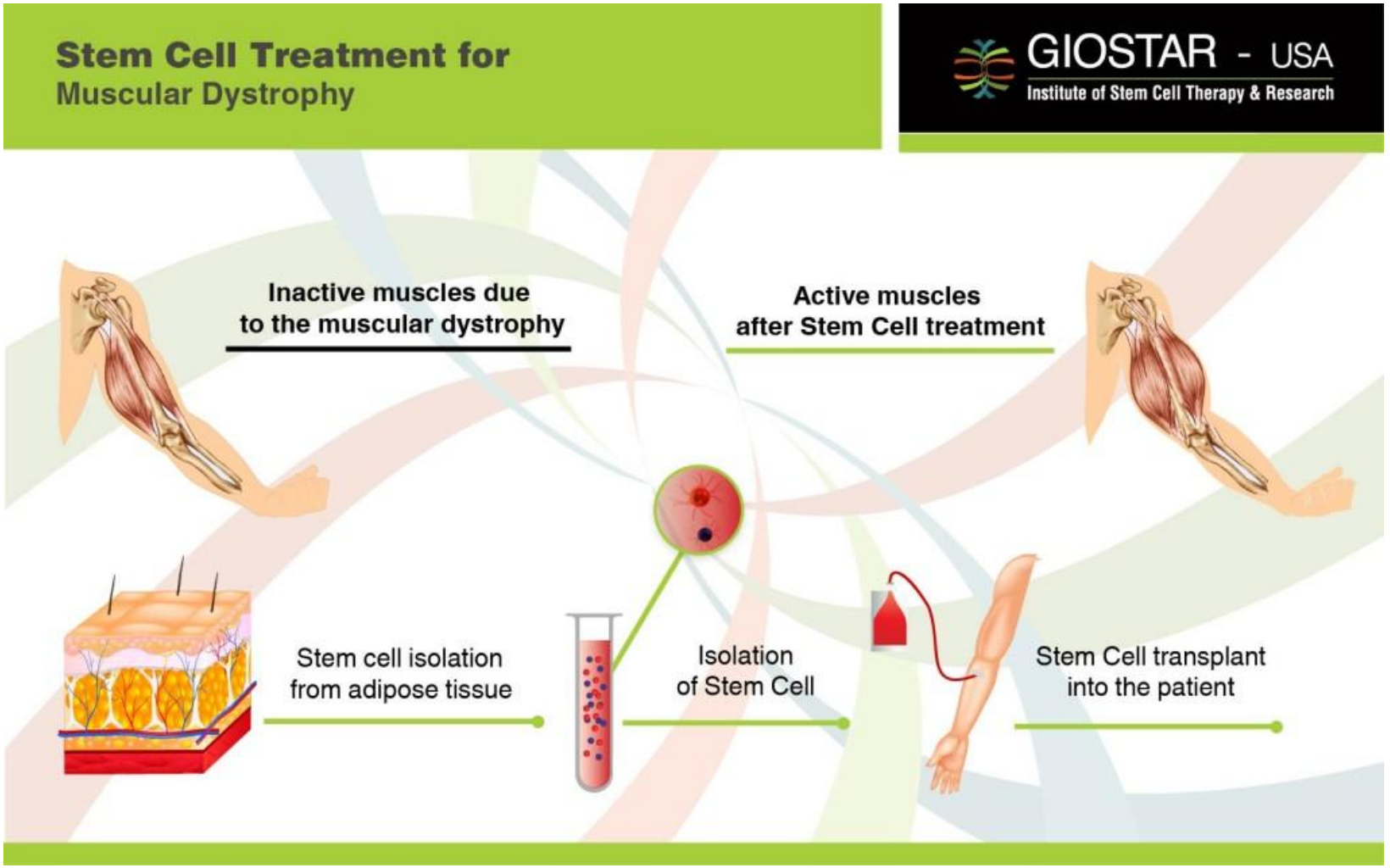
- Symptomatic treatment and physical therapy.

- Corticosteroids to reduce inflammation.

-Trials and experimental treatments:

  -Gene therapy to replace the defective gene and function.

  -Cell therapy: mainly stem cell therapy.



Stem cells → culture → replacement of the faulty gene → differentiation into myogenic cells → patient's reinfusion

Caleb Sizemore's story

Instagram

Entrar Registrarte



mdateammomentum y **caleb.sizemore**

mdateammomentum 70 sem

🌟 Thank you so much to @caleb.sizemore and his family/friends who helped raise $3,000 for @mdaorg in support of finding a cure for Muscular Dystrophy! 🌟 Your generosity means the world to all those affected by this disease. As someone with a mild form of Duchenne Muscular Dystrophy, Caleb knows firsthand the challenges of living with a rare disease. It can be mentally exhausting to manage a physical disability, but your support gives him and others hope for a brighter future. Together, we can make a difference in the fight against Muscular Dystrophy. 💪💙

83 Me gusta

8 de mayo de 2024

Entra para indicar que te gusta o comentar.